

EDUGENIE — SOLUTION REQUIREMENTS

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FUNCTIONAL REQUIREMENTS

- FR-1: User can ask any question and get a concise answer
- FR-2: User can request a simplified explanation of any topic
- FR-3: User can paste text and get a summarized version
- FR-4: User can generate 3 MCQs from any topic or passage
- FR-5: User can request a beginner-to-advanced learning path
- FR-6: All results display in a clean output box on the page
- FR-7: Each feature works independently with its own input field
- FR-8: Quiz output is structured as valid JSON

NON-FUNCTIONAL REQUIREMENTS

- NFR-1: Each response returns in under 15 seconds (average 8–10s)
- NFR-2: Works on any modern browser (Chrome, Edge, Firefox)
- NFR-3: No authentication required (instant access)
- NFR-4: API key stored securely in .env (never in code)
- NFR-5: Graceful error handling for API failures and rate limits
- NFR-6: Clean, distraction-free interface with no ads or trackers
- NFR-7: Mobile responsive (works on phones via local network)

CONSTRAINTS

- Must use FastAPI for the backend (per TN Skills specification)
- Must use Google Gemini (gemini-3.8-flash)
- Must use Jinja2 templates for rendering the UI
- Runs on Windows with Python 3.10+
- Zero cost (free Gemini tier)